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Title: AaaS Labs - Automated Attack and Security Labs

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Introduction

AaaS Labs is an AI-powered cybersecurity assessment and workflow automation system designed to improve security analysis for software repositories and web applications. The system uses Artificial Intelligence and automated security scanning to identify potential vulnerabilities, security risks, and generate actionable recommendations for users.

Problem Definition

Modern software development relies on multiple disconnected security tools, requiring developers to manually run scanners, interpret raw output, and compile findings into reports.

- 1.Developers lack a unified platform to scan both source code repositories and live web domains for vulnerabilities in a single workflow.
- 2.Existing scanning tools (Nmap, Nikto, SQLMap, etc.) operate independently, requiring manual aggregation and correlation of results across multiple outputs.
- 3.Interpreting raw vulnerability scan results and mapping them to standards like OWASP Top 10 and SANS 25 requires specialized security expertise.
- 4.Manual report generation and distribution of findings to stakeholders via email, Slack, or issue trackers is time-consuming and error-prone.
- 5.Lack of automation and AI assistance makes vulnerability remediation slower, increasing the window of exposure to potential attacks.

Objectives

- 1.Develop a unified, no-code platform for orchestrating security assessments of source code repositories and live web domains.
- 2.Integrate multiple scanning tools (Nmap, Gobuster, Nikto, SQLMap, WPScan) into a single automated pipeline via a dedicated scanner microservice.
- 3.Leverage AI (Gemini) to analyze code and scan results, generating security observations and remediation recommendations.
- 4.Automatically generate visual remediation flowcharts and professional PDF reports mapped to OWASP Top 10 and SANS Top 25.
- 5.Enable automated distribution of findings through Slack, GitHub Issues, and Email notifications.

Methodology



Tools used

FRONTEND	BACKEND	SECURITY SCANNING TOOLS	AI & INTEGRATION	DEVOPS & DEPLOYMENT
React.js	Node.js	Simgrep	OpenAI API (GPT-4)	Docker
Bootstrap 5	Express.js	TruffleHog	Google Gemini API	GitHub
Tailwind CSS	Python	OWASP ZAP		GitHub Actions
		Nuclei		
ADDITIONAL TOOLS				
Git	Postman	VS Code	Jira	Notion
				Figma
Built for Security. Modern tools, seamless integration, and automation to deliver reliable security assessments and actionable results.				

Results and Discussions



Conclusions

AaaS Labs successfully demonstrates that repository scanning, domain assessment, AI-driven analysis, and automated reporting can be unified into a single no-code workflow platform. By integrating tools like Nmap, Gobuster, Nikto, SQLMap, and WPScan with Gemini-powered insights and flowchart generation, the system reduces manual effort, improves consistency, and makes vulnerability assessment more accessible to developers and security teams alike.

Outcome of the work

A fully functional cybersecurity orchestration platform with a React-based no-code workflow builder, a Node.js orchestration backend, and a Django-based scanning microservice integrating Nmap, Gobuster, Nikto, SQLMap, and WPScan. The platform supports GitHub OAuth login, AI-powered repository analysis via Gemini, automated Mermaid-based remediation flowcharts, and professional PDF reports mapped to OWASP Top 10 and SANS Top 25, delivered via Slack, GitHub Issues, and Email.

References

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Mentor

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